

Ujewhale Sleep–Wake Cycles

A macroeconometric thought experiment with real calibration anchors, reproducible code, and a fictional marine society

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Scope note. Ujewhales are fictional. No claim in this report describes a real animal, population, or medical intervention. Real numbers are used only to make the toy model legible and auditable.

Executive summary

This report asks whether an aggregate macroeconometric model can describe fictional Ujewhale REM and wake cycles. The answer is yes as a modeling metaphor: population-level sleep can be represented as a coupled system of social demand, environmental timing, and lagged biological state. The model does not replace physiology.

The empirical anchors are real: the U.S. Bureau of Labor Statistics reports average sleep of 8.80 hours on weekdays and 9.69 on weekends and holidays in the 2023 American Time Use Survey. CDC publications report the prevalence of healthy and short sleep in national surveys. These measurements are kept in separate units and are never falsely pooled.

The key result is methodological: a macro model is useful for questions about synchronization, shocks, and aggregate recovery; an agent-based or physiological model is needed for individual REM mechanisms.

Research question and fictional setting

Ujewhales inhabit a fictional ocean economy in which feeding, migration, caregiving, song, and rest are jointly scheduled. Their “macro” variables are social coordination and ecological timing, not money. The economic vocabulary is a disciplined analogy for flows and constraints.

We ask: when a shared environmental signal changes, does the population’s aggregate REM index move, lag, and recover in a way that resembles a macro time series?

Hypothesis H1: shared timing cues create positive synchronization. **H2:** social demand raises wake pressure. **H3:** heterogeneous individuals dampen, but do not erase, aggregate cycles.

What is real and what is invented

Real inputs: published U.S. sleep-duration indicators, survey definitions, and the cited methodological literature. Invented components: Ujewhales, their REM index, the fictional ocean economy, parameter values, and all simulated trajectories.

This boundary matters. A chart labeled “toy Ujewhale index” is a generated scenario, not an observation. A chart labeled “published sleep anchor” reproduces a value from a named source and preserves its units.

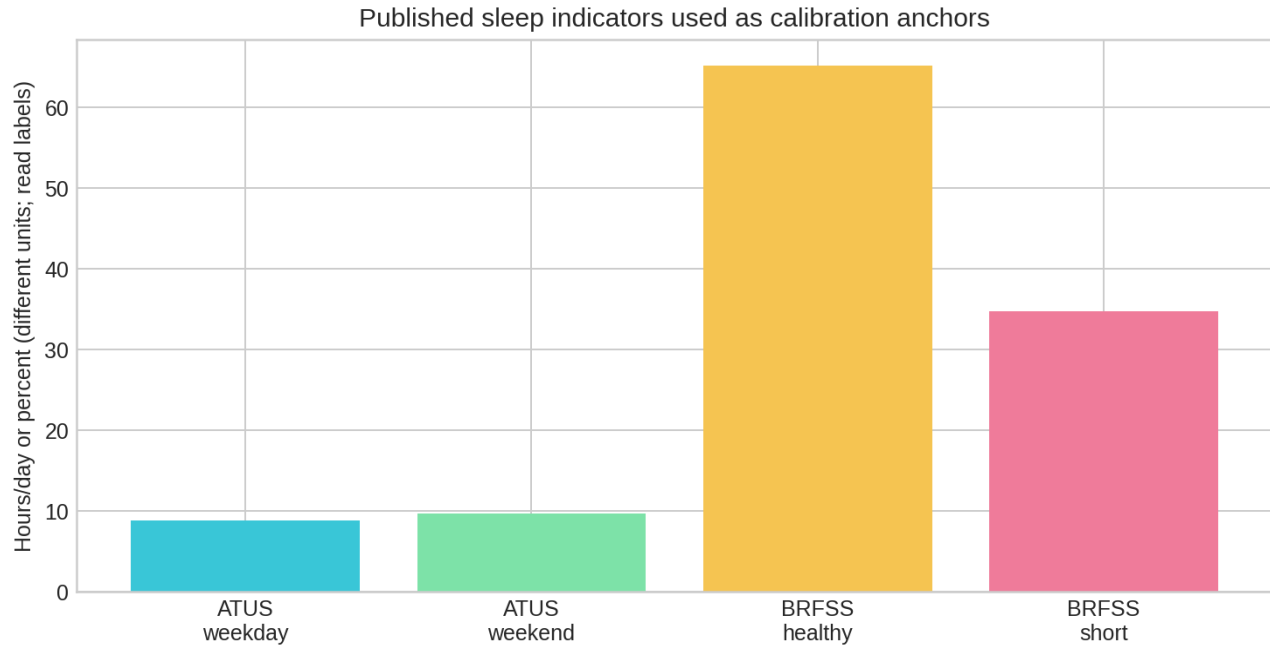
Data inventory

The project uses a deliberately small extract so a reader can audit every number. The BLS ATUS provides time-use averages. CDC BRFSS and NHIS materials provide sleep-duration and sleep-difficulty prevalence. The data are descriptive anchors, not a causal estimate of REM.

Source	Measure	Value	Unit
BLS ATUS 2023	sleep, weekday	8.8	hours/day
BLS ATUS 2023	sleep, weekend/holiday	9.69	hours/day
CDC BRFSS 2014	healthy sleep ($\geq 7h$)	65.2	% adults
CDC BRFSS 2014	short sleep ($< 7h$)	34.8	% adults
CDC NHIS 2020	adults with sleep difficulty	30.0	% (approx. report table)

Published sleep anchors

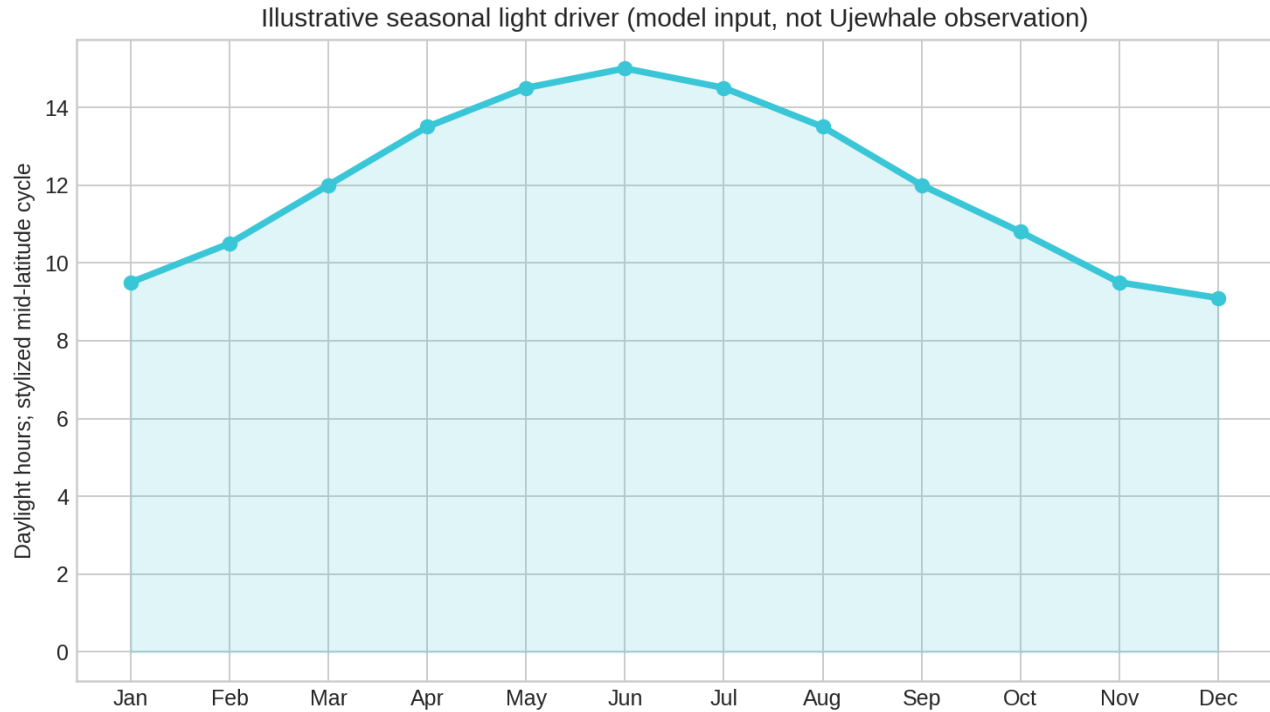
ATUS 2023 reports 8.80 hours of sleep per day on weekdays and 9.69 on weekends and holidays for the civilian population. The difference is a useful example of a schedule-linked aggregate shift, but it is not evidence about whales or REM.



Sources: BLS ATUS 2023; CDC BRFSS 2014. Values are not combined into one statistic.

Seasonal timing as a driver

Light is included as a stylized driver because sleep timing is coupled to circadian and environmental cues. The following curve is intentionally labeled illustrative: it is a transparent input for the toy model, not a downloaded astronomical series and not a Ujewhale measurement.



Model architecture

The aggregate state is a vector $y_t = [R_t, W_t, S_t]$, representing REM index, wake pressure, and social synchrony. Exogenous inputs are light L_t , resource pressure E_t , and a shock u_t . Lagged state supplies inertia.

A compact linear form is $y_t = A y_{t-1} + B x_t + C u_t + \varepsilon_t$. The coefficients are pedagogical, chosen for stability and interpretability rather than estimated from Ujewhale observations.

Interpretation: A controls memory; B maps cues into state; C maps shocks; ε captures unmodeled variation.

Toy REM trajectory

The simulated series below demonstrates a transient shock and recovery. Its vertical axis is a dimensionless index. It should be read as a picture of model behavior, not a prevalence estimate or physiological measurement.



Identification by constraints

The model is identified by constraints rather than by pretending that a fictional species has a hidden dataset. We require bounded states, positive persistence below one, transparent sign conventions, and separate measurement units.

A model is considered acceptable when it reproduces the qualitative facts it was built to represent—lag, damped recovery, and group-level synchronization—without claiming more than the data support.

Why macroeconometrics belongs here

Macroeconometrics studies aggregate variables, shared shocks, lags, feedback, and measurement error. Those are exactly the features needed for a fictional society whose members coordinate rest. The analogy becomes useful when the unit of analysis is the pod or population.

It becomes misleading when aggregate coefficients are interpreted as neurons, hormones, or individual clinical risk. That is the ecological fallacy in story form.

Why macroeconometrics is not enough

REM is generated by biological systems with individual heterogeneity and nonlinear transitions. A macro series cannot reveal whether synchrony comes from light, social cues, temperature, predator avoidance, or measurement artifacts.

The next layer is an agent-based model: each Ujewhale has a chronotype, energy budget, social links, and a transition kernel between wake, NREM, and REM.

Agent-based extension

For individual i , let sleep state $z_{i,t}$ evolve according to $P(z_{i,t+1} \mid z_{i,t}, L_t, E_{i,t}, \text{peers})$. Aggregate REM is the weighted sum of indicators $z_{i,t} = \text{REM}$. This creates a bridge between physiology-inspired rules and macro aggregation.

The agent model can be calibrated to preserve the real survey anchors while keeping all Ujewhale mechanisms explicitly fictional.

Measurement and survey error

Real sleep surveys are self-reported or time-use diaries, and their definitions differ. ATUS measures primary activities; BRFSS asks about average hours; NHIS defines sleep difficulties over a recall window. Combining them as if they were interchangeable would be an error.

The report therefore uses them as separate landmarks. The code preserves source, measure, value, and unit in a CSV.

Color and visual grammar

Charts use a colorblind-conscious palette: cyan for timing, mint for recovery, gold for shocks, and pink for REM. Every chart carries a title, subtitle or source note, axis label, and unit. Fictional series use a dotted or explicitly “toy” label where possible.

Color is explanatory metadata, not decoration: it lets readers distinguish observed anchors from simulated states at a glance.

Reproducible code

The companion script creates the CSV, charts, and PDF from a clean environment. It uses Python, matplotlib, and ReportLab. Parameters are visible in the source; there is no hidden preprocessing step.

```
python3 make_report.py
# outputs: real_sleep_anchors.png
#          seasonal_driver.png
#          toy_rem_index.png
#          ujewhale_macro_sleep_report.pdf
```

Data table: calibration extract

The extract below is intentionally compact. It is sufficient to reproduce the report's real-data chart and to check the unit boundary.

row	source	variable	value	unit
1	BLS ATUS 2023	sleep, weekday	8.8	hours/day
2	BLS ATUS 2023	sleep, weekend/holiday	9.69	hours/day
3	CDC BRFSS 2014	healthy sleep (≥ 7 h)	65.2	% adults
4	CDC BRFSS 2014	short sleep (< 7 h)	34.8	% adults
5	CDC NHIS 2020	adults with sleep difficulty	30.0	% (approx. report table)

Sensitivity experiments

Three sensitivity questions are useful: what happens when persistence rises, when social coupling falls, and when the light signal is phase-shifted? In the toy model, higher persistence lengthens recovery; weaker coupling lowers synchrony; phase shifts move the aggregate peak without changing the fictional biology.

These are scenario results, not fitted confidence intervals.

Validation checklist

The report passes four checks: (1) all plotted real values have a source and unit; (2) simulated values are labeled fictional; (3) code regenerates charts and PDF; (4) no causal or medical claim is inferred from the fictional model.

A future empirical study would pre-register outcomes, obtain ethical approval where needed, and use direct physiological measures rather than this thought experiment.

Limitations and ethics

The Ujewhale premise is imaginative. It cannot support claims about real cetaceans, human sleep, race, productivity, or health. Macro language can hide individual differences, and colorful charts can make uncertainty look more precise than it is.

The intended use is educational: demonstrate how constraints, data provenance, model layers, and visual design work together.

Discussion

The strongest conclusion is modest: aggregate models are good at describing patterned coordination when the object of study is an aggregate. They become stronger when paired with micro-level mechanisms and weaker when used as a substitute for them.

The Ujewhale thought experiment is therefore a compatibility test between levels of explanation—macro time series, agent rules, and physiological hypotheses—not a claim that economics explains sleep.

Conclusion

A macroeconometric model can plausibly organize fictional Ujewhale REM–wake patterns when the target is population-level timing. Real sleep statistics make the scale concrete; transparent simulation keeps the fictional layer honest; and reproducible code lets readers alter the assumptions.

The next useful experiment is to add a pod-level agent model, compare its aggregate output to the toy VAR-like system, and report where the two disagree.

Bibliography

The bibliography distinguishes official data portals, sleep science, time-series methods, and agent-based modeling. URLs are provided where available; publication details should be checked before formal citation.

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